

Which of the following values gives the number of parts per million (ppm) of NaCl in a 0.90% saline solution?

- ☐ A. 900 [7%]
☒ B. 9,000 [47%]
☐ C. 90,000 [13%]
☐ D. 900,000 [31%]

Correct

46% answered correctly

Explanation:

Basis for ppm: $\frac{1,000,000 \text{ parts}}{100 \text{ parts}} = \frac{10,000 \text{ ppm}}{1\%}$

Basis for percent: $\frac{100 \text{ parts}}{1\%}$

Conversion factor: $\frac{10,000 \text{ ppm}}{1\%}$

$0.90\% \text{ NaCl} \times \frac{10,000 \text{ ppm}}{1\%} = 9,000 \text{ ppm NaCl}$

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Similar to the concept of a *percent* (a fraction out of a basis of 100), **parts per million (ppm)** is a fraction out of a **basis of 1,000,000**. For example, 1 ppm means 1 unit in 1,000,000 units.

$$\frac{1}{1,000,000} = \frac{\text{ppm}}{1,000,000}$$

Because the units involved in the fraction are the same (or can be converted to an equivalent by a conversion factor), the units cancel out and ppm (like percent) becomes a **dimensionless quantity**. A fraction correctly expresses ppm as long as the numerator is one million times smaller than the denominator.

For solutions, the density of water (1 g/mL) can be used to relate the mass of water to its volume. Accordingly, 1 L (1,000 mL) of water has a mass of 1 kg (1,000 g). This relationship shows that 1 ppm in an aqueous solution is **equivalent to 1 mg/L**.

$$\frac{1}{1,000,000} = \frac{\text{mg solute}}{1,000,000 \text{ g water}} = \frac{1}{1,000,000} = \frac{\text{mg solute}}{1,000,000 \text{ g water}}$$

The relationship between percent and ppm is a matter of scale because the basis for ppm is 10,000 times larger than the basis for percent ($1,000,000 \div 100 = 10,000$). As a result **1% = 10,000 ppm**. Using this equality as a conversion factor, the equivalent ppm concentration for the 0.90% aqueous NaCl solution can be determined by multiplying the percentage by the conversion factor: $0.90\% \times 10,000 \text{ ppm/percent} = 9,000 \text{ ppm}$.

Alternately, the conversion can be performed more rigorously by setting up an equation involving equivalent ratios and solving for the unknown quantity in the ppm ratio:

$$\frac{x}{1,000,000} = \frac{\text{ppm}}{1,000,000} = \frac{\text{NaCl}}{1,000,000} = \frac{0.90}{100}$$

Therefore, a 0.90% NaCl solution is equivalent to a concentration of 9,000 ppm.

(Choice A) A value of 900 is the equivalent concentration in *parts per one hundred thousand*, not parts per million.

(Choice C) A value of 90,000 is the equivalent concentration in *parts per ten million*, not parts per million.

(Choice D) A value of 900,000 is the equivalent concentration in *parts per one hundred million*, not parts per million.

Educational objective:

Parts per million (ppm) is a dimensionless quantity used to express concentrations as a fraction out of a basis of 1,000,000. In aqueous solutions, 1 ppm = 1 mg/L, and for concentrations expressed as a percentage, 1% = 10,000 ppm.

Time spent: 0 sec

Subject:
General Chemistry

QID:401980

System:
Solutions and Electrochemistry